LendSuri: A Cybersecurity-Focused RAG Chatbot for SEC Regulatory Record Retrieval on Philippine Online Lending Platforms

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***Abstract*— Online lending platforms have made digital credit easier to access in the Philippines, but the same digital environment is also used by fraudulent websites, fake social media pages, unauthorized platforms, and entities that impersonate legitimate lending companies. Filipino users may therefore have difficulty determining what official Securities and Exchange Commission (SEC) records say about a specific Online Lending Platform (OLP). This proposal presents LendSuri, a Retrieval-Augmented Generation (RAG)-based chatbot that retrieves relevant SEC records before generating an answer. Its proposed knowledge base will contain a dated snapshot of the SEC List of Recorded Online Lending Platforms, historical OLP records for comparison, recent SEC advisories and regulatory actions, current OLP regulatory and disclosure guidance, and SEC-specific complaint procedures. The chatbot will explain the available regulatory evidence rather than independently declaring a platform safe, legitimate, or legally authorized. A standard RAG pipeline without language-model fine-tuning is proposed. Future evaluation will measure retrieval accuracy, answer correctness, groundedness, and proper handling of questions that cannot be supported by the available knowledge base.**

Keywords*—*Retrieval-Augmented Generation, RAG, cybersecurity, online lending platforms, SEC, digital lending, regulatory records, large language model

I. SELECTED UN SDG AND JUSTIFICATION

The selected Sustainable Development Goal is SDG 16: Peace, Justice and Strong Institutions. The most relevant area is Target 16.10, which calls for public access to information in accordance with national legislation and international agreements [1]. Target 16.a also provides supporting context because it concerns strengthening relevant national institutions and their capacity to prevent violence and combat crime.

The project remains within the cybersecurity domain because the information problem occurs in a digital environment where users may encounter fraudulent lending websites, fake social media pages, unauthorized platforms, impersonation of legitimate companies, and misuse of regulatory information. In March 2026, the Department of Information and Communications Technology, National Privacy Commission, and SEC jointly reported concerns involving harassment, intimidation, public shaming, and unlawful use of personal data by OLPs [4]. The SEC's 2026 advisories also include unrecorded and unauthorized OLPs, fraudulent platforms, fake Facebook pages, unauthorized websites, and cases involving misuse of legitimate company identities or regulatory information [5], [8].

The issue is relevant to Filipino users because digital lending is already part of the Philippine credit market. TransUnion's 2025 Credit Perception Index reported that 23% of its general-population sample used money lending apps [2]. In addition, the Presidential Anti-Organized Crime Commission reported receiving 47,446 complaints involving allegedly abusive behavior by companies operating online lending applications between August 2024 and January 2026 [3]. A 2025 Bangko Sentral ng Pilipinas discussion paper also examined consumer credit choice in the Philippine digital credit market [7].

A RAG-based chatbot can support this information need by retrieving authoritative regulatory evidence before the language model generates an explanation. The system therefore improves access to reliable information while keeping the SEC, rather than the language model, as the authority behind the answer.

II. NARROW PROBLEM STATEMENT

The broad domain of the proposed study is cybersecurity, narrowed specifically to cyber-enabled deception and regulatory-information uncertainty involving Online Lending Platforms (OLPs) in the Philippines. The intended users are Filipino consumers who encounter or consider using an online lending application, website, social media page, or other related digital lending platform.

The specific problem is that users may have difficulty interpreting what official SEC evidence says about a particular OLP because relevant information may be distributed across the SEC List of Recorded Online Lending Platforms, later SEC advisories and regulatory actions, and earlier historical records [13], [5], [8], [14]. A platform's appearance in an SEC registry at one point in time should not automatically be treated as proof of its current regulatory status when newer SEC evidence is available.

LendSuri will focus on one narrow information need: retrieving and explaining available SEC regulatory evidence for a named Online Lending Platform in the Philippines. The chatbot will check whether the platform appears in the project's dated snapshot of the SEC List of Recorded Online Lending Platforms [13], identify its associated lending or financing company when available, and retrieve any newer SEC advisory or regulatory action involving the platform or

company [5], [8]. A user may enter the name of an OLP and ask whether the platform appears in the SEC records contained in the system, which lending or financing company is associated with it when that information is available, and whether a recent SEC advisory or regulatory action involving the platform or associated company exists.

The study will maintain a clear distinction between a registered lending or financing company and an Online Lending Platform operated or disclosed by that company. This distinction is important because a company's SEC registration and Certificate of Authority concern the lending or financing entity, while the OLP itself is subject to separate disclosure and recording requirements. In July 2026, the SEC issued Memorandum Circular No. 20, Series of 2026, which introduced updated prudential, disclosure, and market-conduct requirements and lifted the previous moratorium on new OLPs [6]. The SEC subsequently issued the 2026 OLP Disclosure Guide, which specifically covers the disclosure and recording of Online Lending Platforms through the SEC iMessage Portal [10].

LendSuri will therefore not simply label an OLP as "registered" or "unregistered" based on a single name match. Instead, it will retrieve the available SEC evidence contained in the knowledge base and explain what that evidence indicates. If the SEC materials identify the platform as unrecorded, unauthorized, or associated with an entity operating without the required authority, the chatbot will present that evidence together with its source and publication date. If sufficient evidence cannot be found, the chatbot will state that the available knowledge base does not contain enough information to support a conclusion.

The second and only follow-up function of LendSuri will be SEC-specific reporting guidance. Since November 4, 2024, complaints involving registered financing and lending companies and their OLPs are filed through the SEC iMessage Portal. For individuals or companies conducting lending activities without a Certificate of Authority, the SEC directs complaints or information to its Enforcement and Investor Protection Department [9].

The chatbot will not recommend lenders, compare loan products, evaluate interest rates, determine whether a loan is financially suitable, guarantee that an OLP is safe or legitimate, investigate criminal activity, determine criminal liability, recover lost funds, or provide general legal or financial advice. It will also not function as a general government-agency routing chatbot. Complaint guidance will remain limited to procedures officially provided by the SEC. These boundaries keep the project focused on one manageable cybersecurity-related information problem.

III. PRIMARY RESEARCH QUESTION

The study will be guided by one primary research question:

"How effectively can a standard RAG-based chatbot retrieve and provide accurate, grounded explanations of available SEC regulatory records and applicable SEC guidance for Online Lending Platforms in the Philippines?"

The question is focused because it is limited to one type of digital platform, one main regulatory authority, and one information need. It is also experimentally measurable. Effectiveness will be evaluated through retrieval accuracy, answer correctness, groundedness, and proper handling of unavailable information.

The phrase "available SEC regulatory records" is intentional. The system will not independently establish the current legal status or safety of every OLP. Instead, it will retrieve and explain evidence contained in a defined knowledge-base snapshot. When multiple records exist for the same OLP or associated company, the system should preserve their dates and explain relevant newer evidence rather than treating one record as automatic proof of current authorization.

IV. INITIAL KNOWLEDGE-BASE DATA SOURCES

The proposed knowledge base will rely primarily on publicly accessible information published by the Securities and Exchange Commission of the Philippines. Using the SEC as the main regulatory authority keeps the corpus focused and reduces conflicts between unrelated sources. A documented knowledge-base cutoff date will be maintained so that the experiment does not imply access to regulatory information published after the dataset was prepared.

A. Recent SEC Online Lending Platform Regulatory Records

The primary positive-registry source will be the SEC's List of Recorded Online Lending Platforms [13]. For experimental reproducibility, the study will use a researcher-prepared dated snapshot of the SEC registry rather than relying only on a live webpage that may change during the evaluation period. The snapshot records the OLP name and its associated lending or financing company based on the available SEC registry information. Inclusion in the snapshot will be treated as evidence that the platform appeared in the SEC registry at the documented snapshot date, not as a permanent guarantee of its future regulatory status.

The SEC itself advises borrowers to verify both the lending or financing company and the Online Lending Platform through its official records before transacting [5].

Recent SEC advisories will provide platform-specific regulatory evidence, including cases involving unrecorded or unauthorized OLPs, mobile applications, websites, social media pages, and other digital lending operations [5], [8]. A 2025 Bangko Sentral ng Pilipinas discussion paper will be used only as supporting research context because it examined the Philippine digital credit market and regulatory data concerning OLPs and lending or financing companies [7]. The BSP study will not replace the SEC as the regulatory authority. For implementation, each SEC record may be stored with information such as the OLP name, associated company, type of regulatory record, publication date, regulatory description, and official source URL so that LendSuri can retrieve and explain the most relevant evidence for a specific platform.

B. Historical SEC OLP Records

The SEC's List of Recorded Online Lending Platforms updated November 29, 2021 will be retained as a historical baseline [14]. This historical source will

support comparison between earlier and newer OLP records. An appearance in the 2021 list will not be treated as proof of current authorization because later SEC registry information, advisories, suspensions, revocations, or other regulatory actions may affect how the earlier record should be interpreted.

C. Current SEC Regulatory Framework and OLP Disclosure Guidance

The knowledge base will also include SEC Memorandum Circular No. 20, Series of 2026 [6] and the 2026 OLP Disclosure Guide [10]. MC No. 20 introduced updated prudential, disclosure, and market-conduct requirements for financing and lending companies and lifted the previous moratorium on new Online Lending Platforms. The OLP Disclosure Guide provides procedures for disclosure and recording of Online Lending Platforms through the SEC iMessage Portal. These sources provide current regulatory context for interpreting registry information.

D. SEC Advisories and Regulatory Actions

SEC advisories and regulatory actions will provide later platform-specific evidence involving unrecorded or unauthorized OLPs, fraudulent platforms, impersonation, unauthorized websites or social media pages, suspensions, revocations, and other regulatory actions [5], [8]. These records are important because newer SEC evidence may change how an earlier registry entry should be interpreted. Each record will preserve the platform or associated company, type of regulatory action, publication date, and official source. LendSuri will not infer a regulatory conclusion beyond what the SEC source states.

E. SEC Complaint and iMessage Guidance

The third source will consist of the SEC's official complaint procedures. The SEC states that beginning November 4, 2024, complaints against registered financing and lending companies and their Online Lending Applications or Platforms should be filed through the SEC iMessage Portal [9]. For individuals or companies conducting lending activities without a Certificate of Authority, the SEC directs complaints or information to its Enforcement and Investor Protection Department. This source is directly relevant to questions asking what SEC-specific complaint procedure applies.

To reduce misleading conclusions when multiple records concern the same OLP or associated company, retrieved evidence will retain source dates and evidence type. When relevant records conflict or differ over time, newer SEC advisories, enforcement actions, suspensions, or revocations will be presented together with earlier registry information. The system will not treat an older positive registry record as proof of current authorization, and absence from the available corpus will be treated as insufficient information rather than automatic evidence of authorization or non-authorization.

V. INITIAL EVALUATION DATA PLAN

The evaluation data will remain separate from the RAG knowledge base. A preliminary test set of approximately 30 questions may be developed after the knowledge-base sources have been collected and verified. Most questions will be answerable using the defined SEC corpus, while a smaller number will intentionally require information that is not available in the knowledge base.

Realistic user questions may be derived from cases described in SEC advisories and from the types of information a Filipino consumer would reasonably request about a named OLP. Question types may include whether an OLP appears in an SEC record, which company is associated with it, whether a later regulatory advisory exists, what the most recent available regulatory evidence says, and which SEC complaint procedure applies based on the retrieved evidence.

The evaluation set will include several case types: OLPs appearing in the dated 2026 registry snapshot with no identified later adverse record; platforms represented in both historical and newer records; platforms or associated companies affected by later SEC advisories or regulatory actions; OLPs explicitly identified by the SEC as unrecorded or unauthorized; and platform names for which no sufficient evidence exists in the knowledge base.

The evaluation will also include cases where an earlier SEC regulatory record and a newer SEC advisory or regulatory action concern the same platform or associated company. These questions will test whether the retriever can surface the relevant evidence while preserving document dates and avoiding the assumption that older information automatically represents the current situation.

For every answerable question, an expected answer and one or more supporting source passages will be established in advance from the official records. A small number of questions will intentionally refer to platforms or facts not represented in the knowledge base. For these cases, the expected behavior is to state that sufficient evidence could not be found rather than guessing or automatically describing the platform as authorized, unauthorized, safe, legitimate, or fraudulent.

Because the proposed language model supports Filipino and English, a small portion of the questions may be written in Filipino or natural Filipino-English phrasing. This will check whether the proposed model can serve the intended users without making multilingual performance a separate primary research question.

The future evaluation will consider: retrieval accuracy, whether the expected SEC source appears among the retrieved documents; answer correctness, whether the generated response agrees with the verified expected answer; groundedness, whether the important claims in the response are supported by the retrieved evidence; and unavailable-information handling, whether unsupported questions correctly trigger an insufficient-evidence response rather than a hallucinated conclusion.

VI. POSSIBLE LANGUAGE MODEL AND JUSTIFICATION

The proposed language model is Llama-SEA-LION-v3 8B-IT, an instruction-tuned model developed by AI Singapore for Southeast Asian languages. Its official documentation identifies the model family as based on an 8-billion-parameter architecture with a context length of up to 128K tokens. Filipino and English are among its supported languages [11]. The model is suitable for the proposed task because users may naturally ask questions in English, Filipino, or a mixture of both.

Its expected role is not to provide regulatory knowledge from its pretrained memory. Instead, the RAG pipeline will retrieve the relevant SEC records and place them in the model's context, after which the model will produce a concise explanation grounded in those records. Fine-tuning is not planned for the initial experiment. SEC regulatory information is time-sensitive and may change, making an external knowledge base more practical than encoding such information directly into model weights.

New SEC records can therefore be added to the corpus without retraining the language model. From a practical perspective, an 8B model may require more GPU memory than is consistently available in free or limited Google Colab runtimes. Depending on available resources, local or hosted inference may therefore be considered [11], [12]. The model remains a preliminary choice, not a claim that it is the best language model. Its actual performance on the proposed Filipino and English OLP questions will be evaluated rather than assumed.

VII. GROUP REFLECTION

The group considers the proposed problem sufficiently narrow because LendSuri is not intended to become a general financial, lending, scam-reporting, legal, or cybersecurity assistant. The study deals with one type of digital service, Online Lending Platforms, and relies mainly on one regulatory authority, the Philippine SEC. Its information task is also specific: retrieve and explain available SEC regulatory records and related SEC guidance concerning a named OLP.

The project remains aligned with the group's cybersecurity domain because users may encounter cyber-enabled deception through fake lending applications, fraudulent websites, fake social media pages, impersonation of legitimate companies, unauthorized platforms, and misuse of regulatory information. The chatbot will not detect malware or investigate offenders. Instead, it addresses a narrower cybersecurity information problem by helping users retrieve authoritative regulatory evidence before relying on the identity or claims of a digital lending platform.

The proposed dataset is manageable for a student project because it can be restricted to a dated 2026 snapshot of the SEC OLP registry, the historical 2021 recorded-OLP list, selected SEC advisories and regulatory actions, current regulatory and disclosure guidance, and SEC-specific complaint procedures. The evaluation set can likewise remain small enough for manual verification while still covering current-registry, historical, adverse-regulatory, and unavailable-information cases. RAG is suitable because the relevant evidence is distributed and time-sensitive. An earlier regulatory record and a later SEC advisory may both be relevant to the same user question. Keeping these sources outside the language model allows the knowledge base to be updated without fine-tuning or retraining.

Most importantly, the study investigates a measurable information-retrieval problem rather than simply proposing to build another chatbot. The future experiment will determine whether a standard RAG pipeline can retrieve the correct regulatory evidence and produce grounded explanations without making claims that the SEC sources do not support.

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